

LEMC Experiment Gallery

Fixed Geometry vs. Learned Residuals for Ego-Motion Compensation

Companion document to `main.tex` | Three seeds, PIE + JAAD

August 30, 2026

Headline results (test split, mean $\pm$ std, 3 seeds)

- **Fixed ORB affine:** PIE ADE **60.5 $\pm$ 1.8** px (neutral vs. baseline 60.8); JAAD ADE **62.7 $\pm$ 2.3** px (−25% vs. 83.1).
- **Variance gate** (standing+moving): fixed ORB passes **93.1%** (PIE) and **96.9%** (JAAD); learned LEMC+OBD passes **0.8%**.
- **Learned LEMC** (all variants): worse ADE than baseline; OBD correlation $R^2=0.30$ without physical validity.
- **Intent shortcut:** AUC ordering fixed ORB (0.80) < baseline (0.89) < GT speed (0.93).

Setup. Observation $\tau=16$, horizon $\eta=45$, TTE $\in [30, 60]$, 50% overlap. PIE: 1,773 test windows (augmented multi-agent tracks). JAAD: 295 test windows, ORB from 124 scenes, no OBD at inference. Backbone: linear embed $\rightarrow$ 2-layer GRU $\rightarrow$ trajectory + intent heads.

1 Method

Offline ORB keypoints on background (pedestrian boxes masked) yield a partial affine M per frame pair. Compensation is evaluated at each agent’s image centre—not as a global translation—before GRU prediction. Inference uses precomputed affines and bbox tracks only (no pixels, no speed sensor).

Fixed ORB-affine compensation (sensor-free at inference)

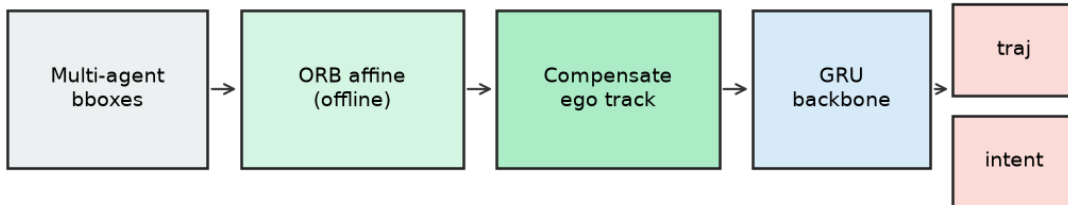


Figure 1: Fixed ORB-affine pipeline. ORB extraction is offline; at inference only track coordinates and stored affines are required.

2 PIE Results

2.1 Main comparison grid

Table 1: PIE test set (mean $\pm$ std, 3 seeds). Best trajectory metrics in bold.

Method	ADE	FDE	ARB	FRB	AUC	F1
Baseline (no compensation)	60.81 $\pm$ 3.31	121.75 $\pm$ 9.87	33.96 $\pm$ 1.42	56.52 $\pm$ 4.25	0.89 $\pm$ 0.01	0.74 $\pm$ 0.03
GT OBD speed (oracle)	66.96 $\pm$ 1.28	136.89 $\pm$ 3.02	36.20 $\pm$ 1.83	59.66 $\pm$ 2.06	0.93$\pm$0.01	0.85$\pm$0.02
LEMC, no auxiliary loss	81.68 $\pm$ 5.13	159.91 $\pm$ 2.39	44.20 $\pm$ 4.35	70.56 $\pm$ 3.22	0.77 $\pm$ 0.05	0.64 $\pm$ 0.01
LEMC + OBD auxiliary	85.77 $\pm$ 11.19	164.25 $\pm$ 11.33	47.36 $\pm$ 8.24	73.56 $\pm$ 6.67	0.80 $\pm$ 0.01	0.60 $\pm$ 0.03
LEMC + ORB regression	81.73 $\pm$ 4.61	163.54 $\pm$ 7.55	43.74 $\pm$ 2.29	71.93 $\pm$ 5.51	0.81 $\pm$ 0.01	0.63 $\pm$ 0.00
Fixed ORB affine	60.49$\pm$1.83	130.28 $\pm$ 1.31	32.34$\pm$1.75	54.49$\pm$0.44	0.80 $\pm$ 0.01	0.60 $\pm$ 0.01

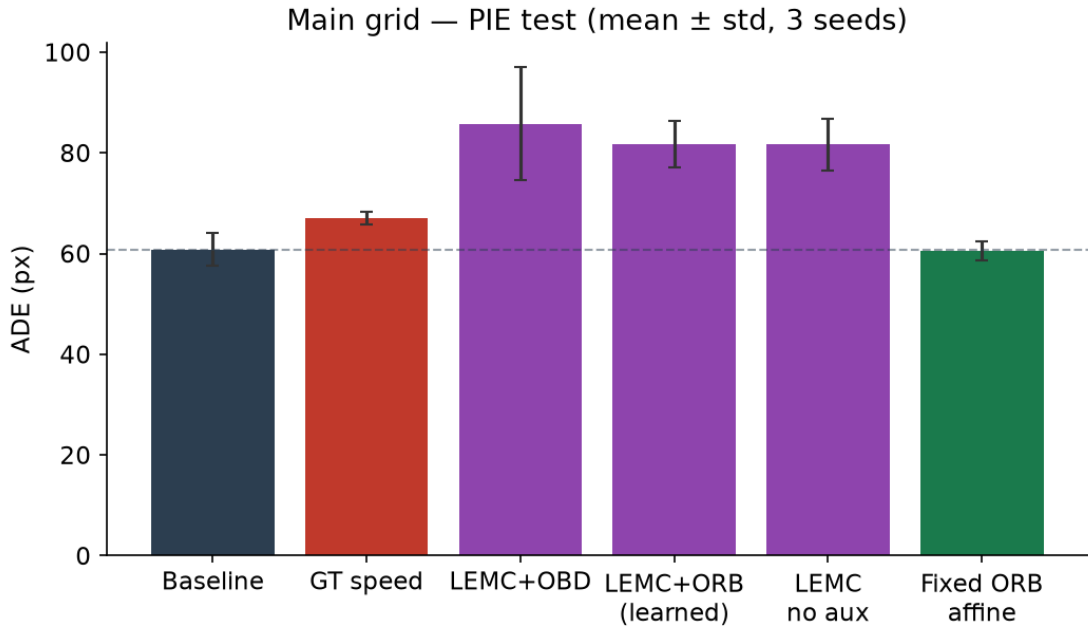


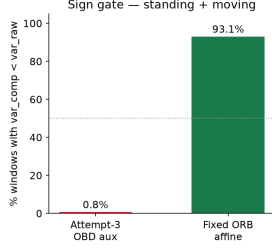
Figure 2: PIE ADE across conditions (3 seeds). Fixed ORB matches baseline; every learned LEMC variant degrades trajectory error.

2.2 Geometric ablation and variance gate

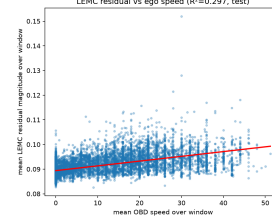
Translation-only ORB (or learning to regress it) fails the standing-pedestrian test: apparent motion from rotation/scale depends on image position.

Table 2: Variance-gate pass rate (% windows with $\text{Var}(\text{comp}) < \text{Var}(\text{raw})$ on standing pedestrians while vehicle moves).

Compensation	Scope	Diagnostic ($n=300$)	Full PIE test
ORB translation only	standing+moving	~ 21	—
Fixed full affine	standing+moving	~ 89	93.1
Learned LEMC + ORB-reg.	standing+moving	—	33.4
LEMC + OBD auxiliary	standing+moving	—	0.8



(a) Sign gate pass rates



(b) LEMC residual vs. OBD ($R^2=0.30$)

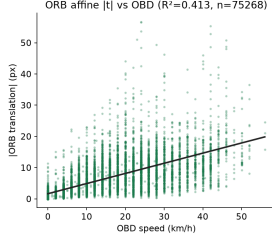
Figure 3: Physical validity vs. statistical correlation. Learned LEMC correlates with OBD speed but fails to flatten standing pedestrians.

3 ORB Validation and Stratified PIE

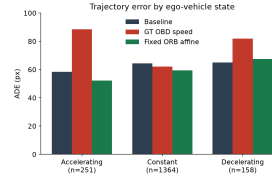
ORB translation magnitude vs. OBD speed: $R^2=0.41$ ($n=75,268$ frame pairs)—a geometric sanity check independent of the predictor. During **acceleration**, GT-speed conditioning yields the worst ADE (88.6 px); fixed ORB is best (52.2 px), matching LIM’s causal-confusion pattern.

Table 3: ADE/FDE (px) by ego-vehicle state, PIE seed 0.

State	n	Baseline	GT speed	Fixed ORB
Accelerating	251	58.5 / 127.8	88.6 / 173.2	52.2 / 122.6
Constant	1364	64.5 / 130.6	62.2 / 120.3	59.4 / 130.1
Decelerating	158	65.0 / 139.8	82.0 / 174.7	67.5 / 151.5



(a) ORB $|t|$ vs. OBD ($R^2=0.41$)



(b) ADE by ego state (seed 0)

Figure 4: ORB validation and stratified breakdown. GT speed helps only under constant velocity; hurts under acceleration.

4 JAAD Cross-Dataset

JAAD has no continuous ego-motion labels—the harder test for sensor-free claims. Fixed ORB reduces ADE by 25%, FDE by 23%, ARB by 22%, and FRB by 24%. Intent AUC remains at chance (~ 0.44 – 0.47); gains are trajectory-only.

Table 4: JAAD test set ($n=295$, 3 seeds).

Method	ADE	FDE	ARB	FRB	AUC	F1
Baseline	83.10 $\pm$ 2.00	156.25 $\pm$ 4.31	44.76 $\pm$ 2.15	71.23 $\pm$ 2.08	0.47	0.90
Fixed ORB	62.68$\pm$2.34	120.11$\pm$2.82	34.80$\pm$2.25	54.17$\pm$0.91	0.44	0.90

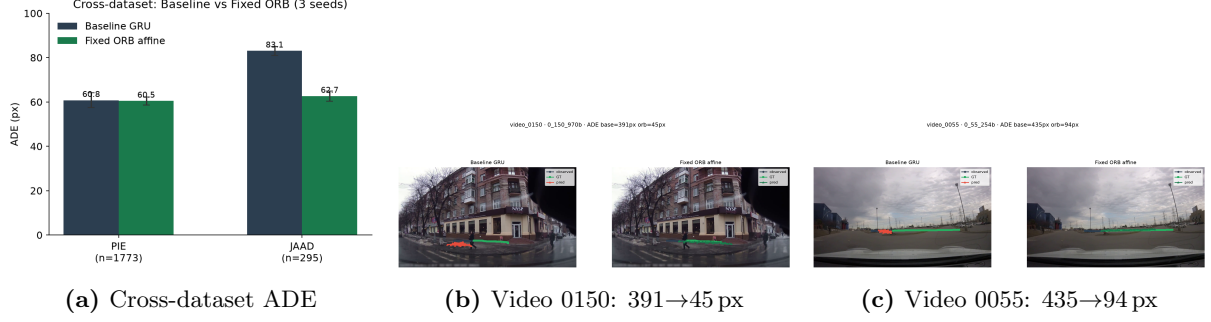


Figure 5: JAAD: PIE-neutral, JAAD-improving. Representative windows where compensation collapses large baseline errors.

5 Intent Shortcut and Qualitative Examples

Intent AUC rises monotonically with ego-motion exposure: fixed ORB (0.80) < uncompensated baseline (0.89) < GT speed (0.93). Each increment injects ego-motion-correlated cues that can inflate apparent intent accuracy.

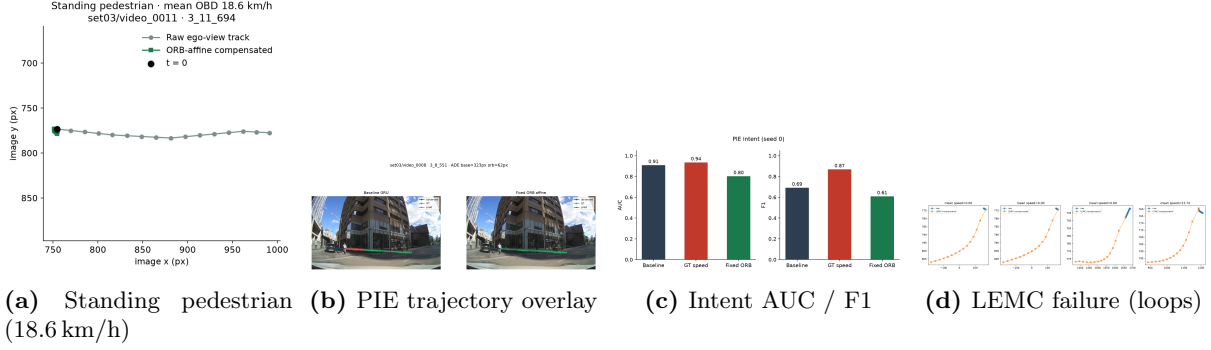


Figure 6: Qualitative summary. (a) ORB flattens a standing pedestrian track; (b) PIE overlay on real frame; (c) intent vs. ego-motion; (d) OBD-supervised LEMC loops instead of flattening.

6 Takeaways

1. **Physical gate before ADE.** Correlation with OBD ($R^2=0.30$) does not imply valid compensation; the variance gate separates the two.
2. **Fixed beats learned.** Position-dependent full-affine ORB passes the gate and matches or beats baseline ADE; learned multi-agent residuals do not.
3. **Dataset-dependent downstream gain.** PIE is trajectory-neutral (-0.5 px); JAAD gains 20 px because raw tracks carry more uncompensated ego drift without OBD.
4. **Intent remains entangled.** Sensor-free geometry helps trajectory; intent metrics still track available ego-motion unless explicitly controlled.

A Figure file index

File (paper/figures/)	Gallery section	Role in <code>main.tex</code>
<code>fig2.architecture.png</code>	§1	Method pipeline
<code>fig_main_grid_ade.png</code>	§2	PIE ADE bar chart
<code>fig_sign_gate.png</code>	§2	Variance gate bars
<code>lemc_obd_scatter_...png</code>	§2	LEMC–OBD scatter
<code>fig4_orb_obd_scatter.png</code>	§3	ORB–OBD validation
<code>fig5_stratified_ade.png</code>	§3	Stratified ADE
<code>cross_dataset_ade.png</code>	§4	PIE vs. JAAD bars
<code>jaad.compare_0.png</code>	§4	JAAD qualitative
<code>jaad.compare_1.png</code>	§4	JAAD qualitative
<code>fig1_teaser.png</code>	§5	Standing pedestrian
<code>pie.compare_0.png</code>	§5	Frame overlay
<code>pie_intent_auc_f1.png</code>	§5	Intent AUC/F1
<code>lemc_qualitative_...png</code>	§5	LEMC failure case

Table 5: All 13 PNG files used in the conference paper. `fig3_qualitative.png` is not included (superseded).

Compile: `cd paper && pdflatex figures_gallery.tex`